

DELTA TEB Series

TEB4810 – 3U 48V/100Ah

Lithium-ion Iron Phosphate (LFP) Battery



This new DELTA 48V battery pack is designed with 100Ah capacity battery cell of lithium-ion iron phosphate chemistry. It provides larger capacity in a compact size of 19" rack-mounted 3U chassis. This pack has RS485 communication and built-in BMS with automatic protection and cell balancing to offer safe and most efficient operation to customers.

The pack application is developed for Telecom equipment power backup. Under normal condition, grid AC power supplies to rectifier module and the Telecom loads and also charge battery pack.

When the AC power failed, rectifier module stop power supply to loads, the battery serves for Telecom equipment to ensure the Telecom equipment operation normally; when the AC power is switched on again, power from rectifier module to Telecom equipment recovered and charge the battery pack back to backup status.

Key Features

- RS485 communication output for monitoring
- Built-in automatic protection for over-charge, over-discharge and over-temperature conditions
- Built-in BMS with Charging current limiting circuit
- Built-in battery optimal control algorithm to manage sleep/wakeup and intermittent charging mode to improve operation efficiency.
- Built-in Cell balancing function
- LED indication of State of Charge and Alarm/Run status
- Compatible with standard Telecom rectifiers
- Maintenance free

Applications

- 3G / 4G / 5G
- Fixed Line
- Datacom



DELTA, TEB4810 – 3U 48V/100Ah

NOMINAL

Voltage	48.0V _{DC}
Nominal Capacity (25°C)	100Ah (capacity is measured under to 0.2C / 25°C @ BOL)

ELECTRICAL

Energy	
- Nominal Energy (25°C, 0.2C)	4800Wh (capacity is measured under to 0.2C / 25°C @ BOL)
- Volumetric Energy Density	184Wh/L
- Gravimetric Energy Density	123Wh/kg
Standard Discharge (25°C)	
- Maximum continuous Current	100A
- Cut-off Voltage	37.5V _{DC}
- LVBD Voltage	41.3V _{DC}
Standard Charge (25°C)	
- Maximum continuous Current	100A
- Recommended Charging Current	50A
- Charge Voltage	Max 53.5 VDC (Recommend: 51.9VDC)
Charge current limit function	10A
Internal impedance	<19mΩ

MECHANICAL

Dimensions (W x H x D)	Without handle: 442 (±1) x 131 (±1) x 450 (±1) mm With handle & terminal: 482 (±1) x 131 (±1) x 491 (±1) mm
Weight	approx. 39.0±2.0kg
Total Cells Quantity in Battery Module	15pcs of cell in series (Prismatic type cell)
Materials of Battery case	Carbon steel with corrosion resistant coating
Cell	LFP 100Ah
Max Quantity of Battery Parallel Connection	15 for one RS-485 and max 30 pcs in system.
Color	RAL9003 (White)

ENVIRONMENTAL

Calendar Life (25°C)	> 10 years
Cycle Life (1C/1C, DOD 100%, 25°C)*	SOH 80%, 5000 cycles
Operating Temperature**	0 to +50 °C (De-rating from 35 °C)
Storage Temperature	0 to +45 °C (Recommended range: 10 to +30 °C)

BMS PARAMETERS

Voltage (Charge)	Cell voltage protection: 3.75V protection (Recover at 3.45V) Total voltage protection: 55.5V protection (Recover at 50.4V)
Voltage (Discharge)	Cell voltage protection: 2.5V protection (Recover at 3.1V) Total voltage protection: 37.5V protection (Recover at 45V)
Current (Charge)	Normal ≤ 100A Charging current limiting function: > 110A (10sec) Charging current limiting function: > 125A (3sec)
Current (Discharge)	Normal ≤ 100A Over current protection 1: > 110A and < 125A (10sec) Over current protection 2: > 125A (3sec) Short circuit protection: ≥ 320A (<1ms)
Temperature (Cell)	Low temperature protection: Charging < 0°C, Discharging < -20°C High temperature protection: Charging > 55°C, Discharging > 60°C
Temperature (PCB)	Temperature protection ≥95°C (Recovery < 75°C)
Cell Balance (Balance)	BMS include Passive balancing circuit and algorithm , (Current: < 100mA)
Dry contact	D.O x 2

STANDARDS

Safety	UN38.3, IEC62619, IEC60730, CE, UL1973
EMC	IEC EN 61000-6-1

ORDERING INFORMATION

BSPMT-GT1151C0US0	BATTERY PACK LI-ION 48.0V 100AH
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* 1C/1C means charge 1C and discharge 1C

** Operation temperature means the ambient temperature. Protection may be triggered if the conditions defined in the specification are exceeded.

All specifications are subject to change without prior notice.

Performance may vary depending on, but not limited to cell usage and application. If cell is used outside specifications, performance will diminish.

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